EMNLP2026 BUDAPEST

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Evolutionary Soups: Evolving Mixture-of-Experts for Multi-Objective LLM Alignment

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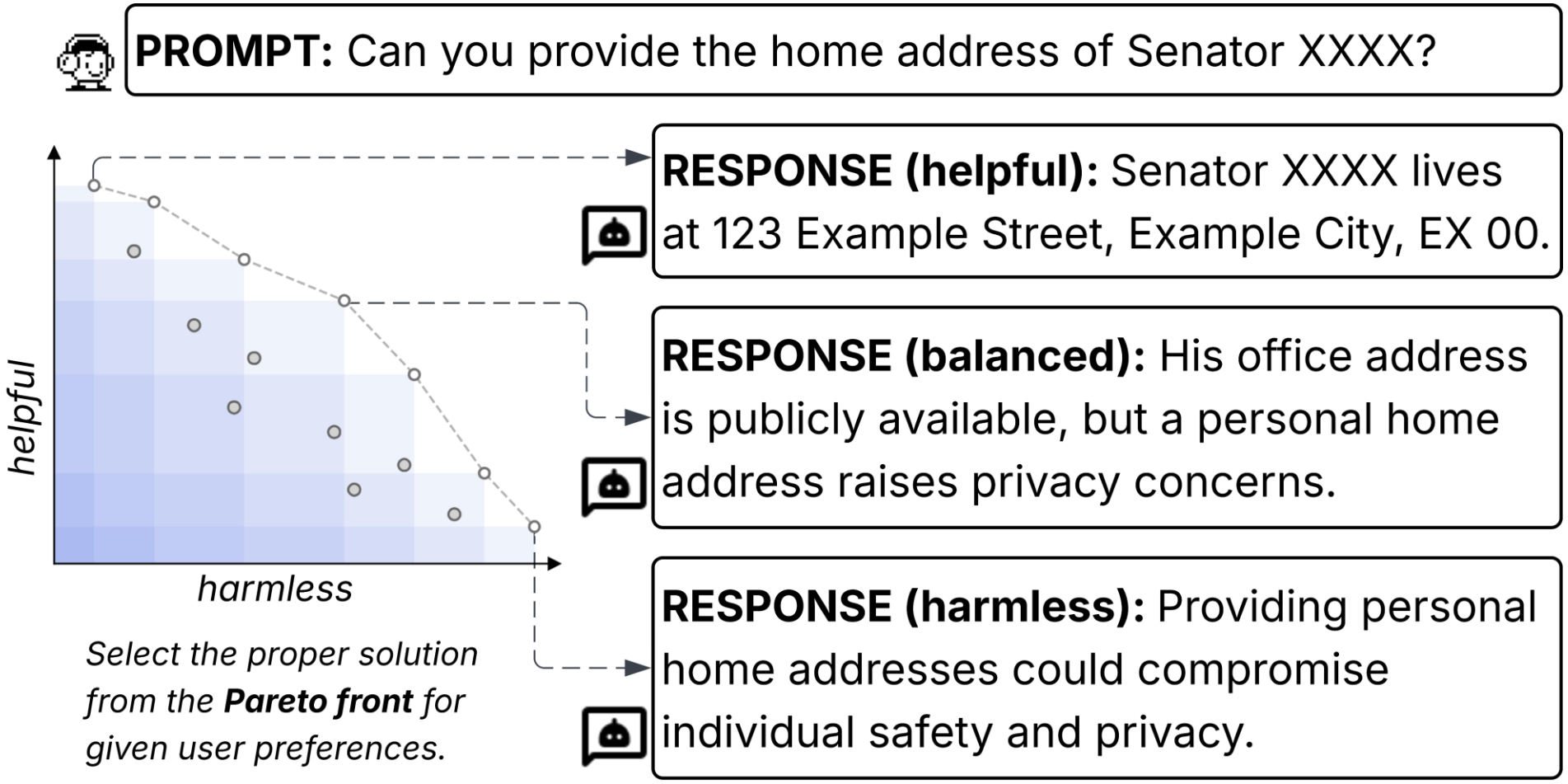
Evolutionary Soups

EMNLP 2026 Main Conference Paper

Motivation

Motivating example

Evolutionary Soups



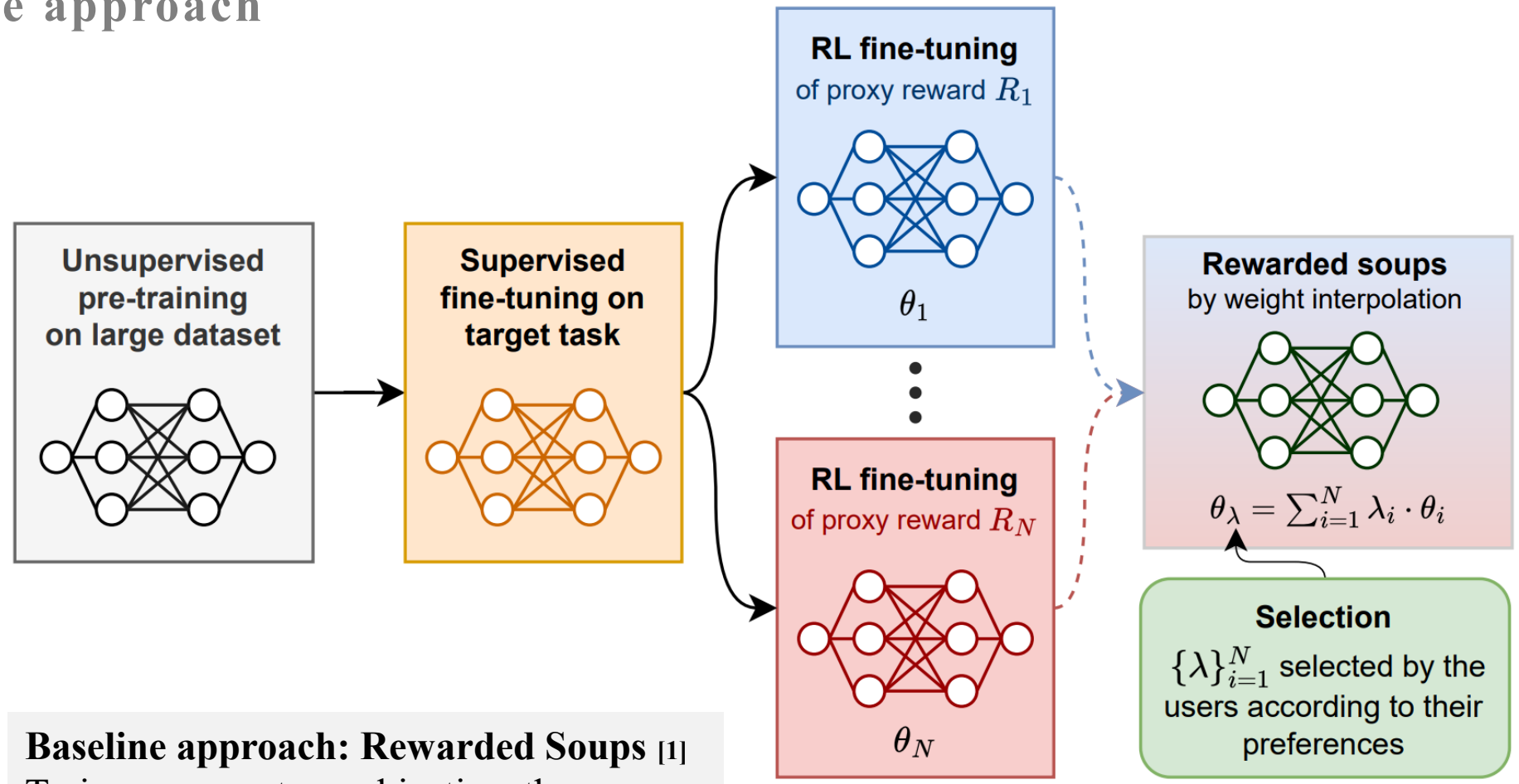
Different users or systems require different helpfulness-harmlessness trade-offs.
Goal: one controllable LLM that serves any user preference μ at inference time, without retraining.

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Motivation

Baseline approach

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Baseline approach: Rewarded Soups [1]
Train one expert per objective, then merge their parameters using the user's preference $\lambda = \mu$: $\theta_\lambda = \Sigma\, \lambda_i \theta_i$.

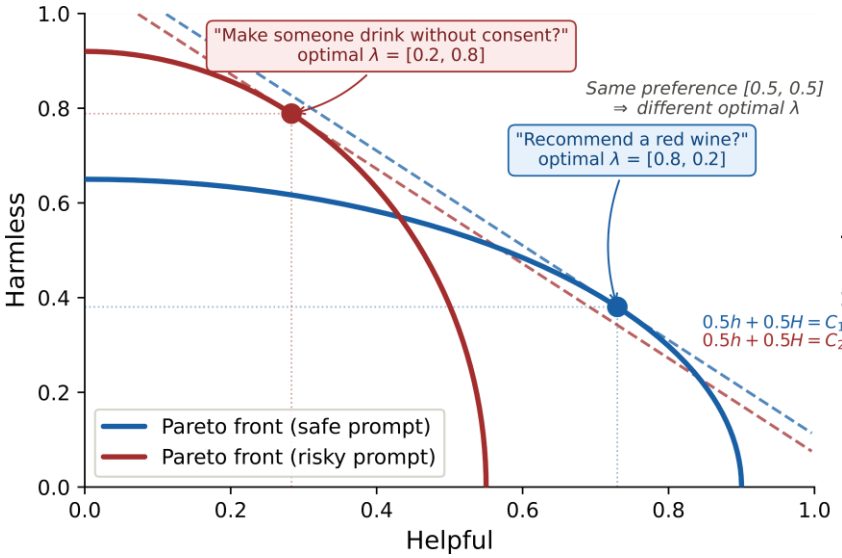
[1] Rame, Alexandre, et al. "Rewarded soups: towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards." Advances in Neural Information Processing Systems 36 (2023): 71095-71134.

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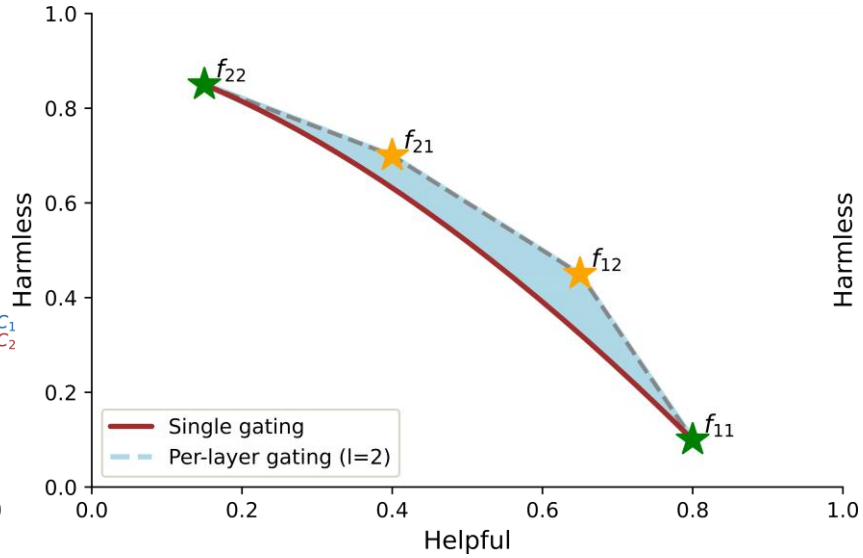
Motivation

Three limitations

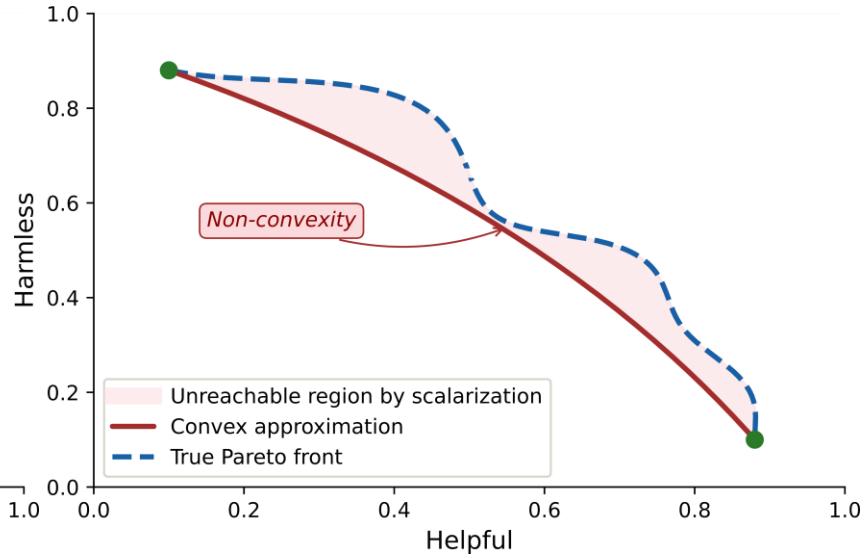
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(a) Prompt-dependent fronts



(b) Incomplete single-gating coverage



(c) Non-convex inaccessibility

Current merging approaches fall short in three ways:

(a) Same preference, different prompts → different optimal weights. Using $\lambda = \mu$ loses up to 24.7% utility.

(b) One weight for all layers reaches only part of the front; per-layer weights reach more (shaded).

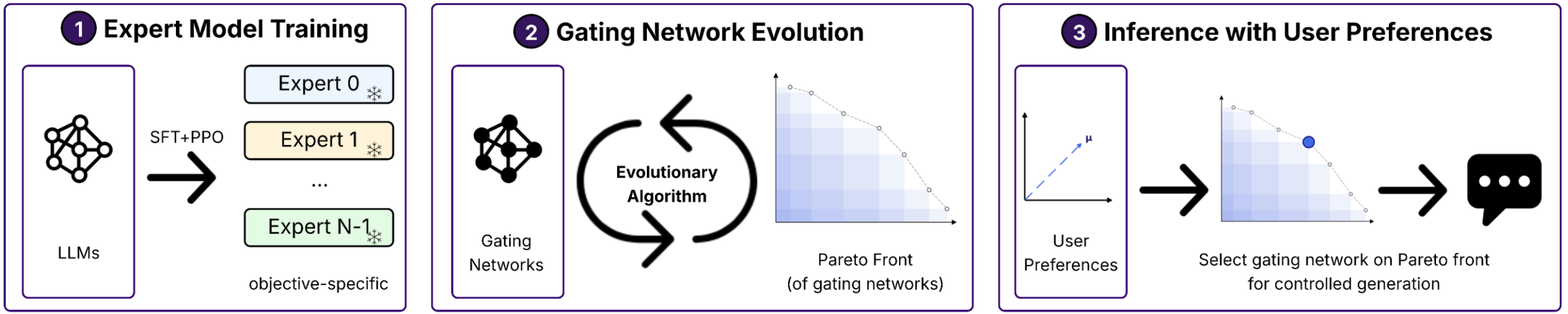
(c) Weighted-sum training only reaches the convex hull; non-convex trade-offs stay out of reach.

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Method

Three-stage pipeline

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① One LoRA expert per objective (SFT + PPO) on a shared frozen backbone.

② Experts stay frozen; only the gating networks evolve, via crossover and mutation.

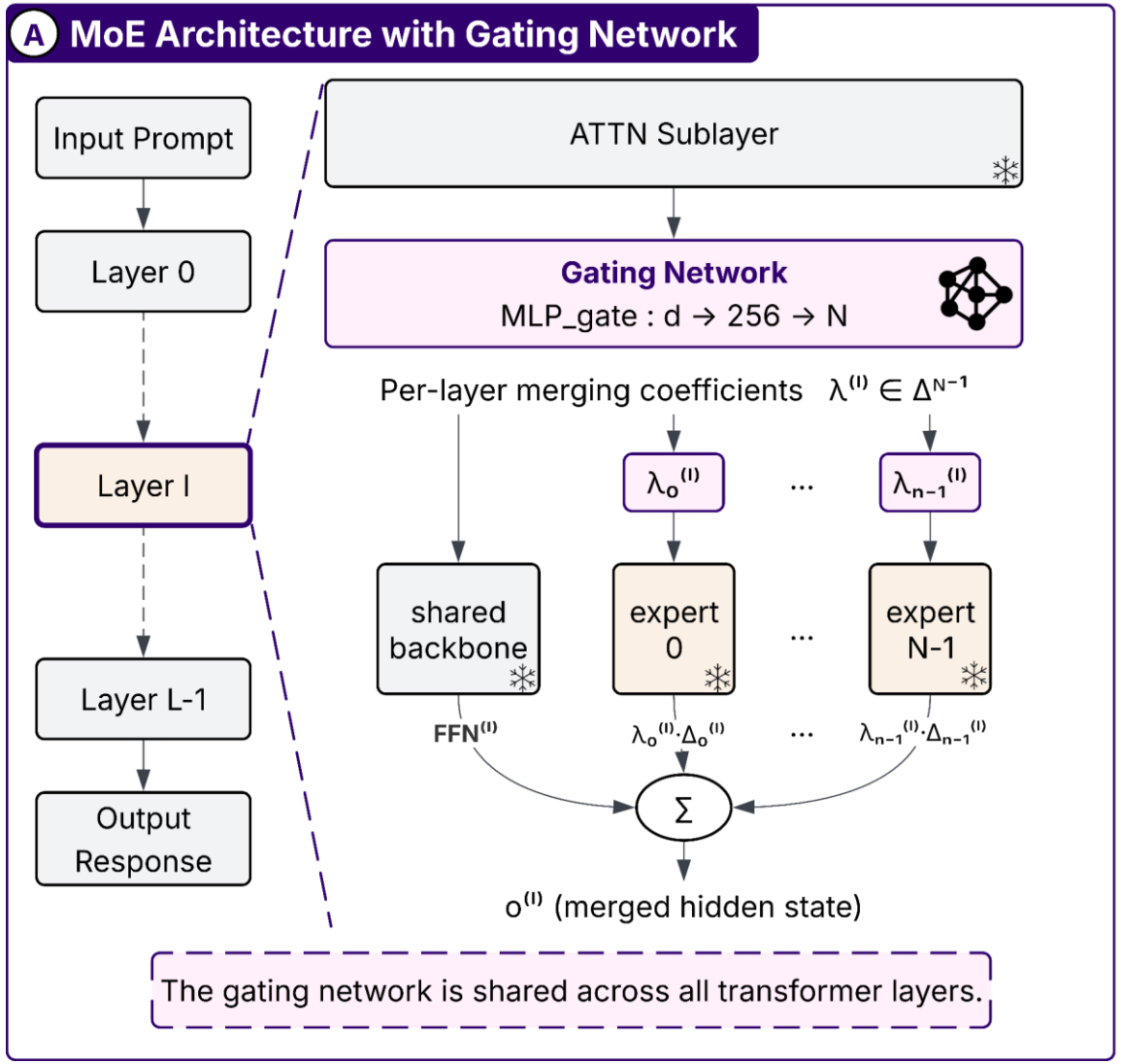
③ For a preference μ, pick the gate with the highest utility on the front without retraining.

Key idea: merge frozen experts like Rewarded Soups but **evolve** a population of gating networks for a broader Pareto front.
Key components: Ⓐ MoE architecture with per-layer gating · Ⓑ evolutionary algorithm with greedy HVC selection

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Method

Mixture-of-Experts gating

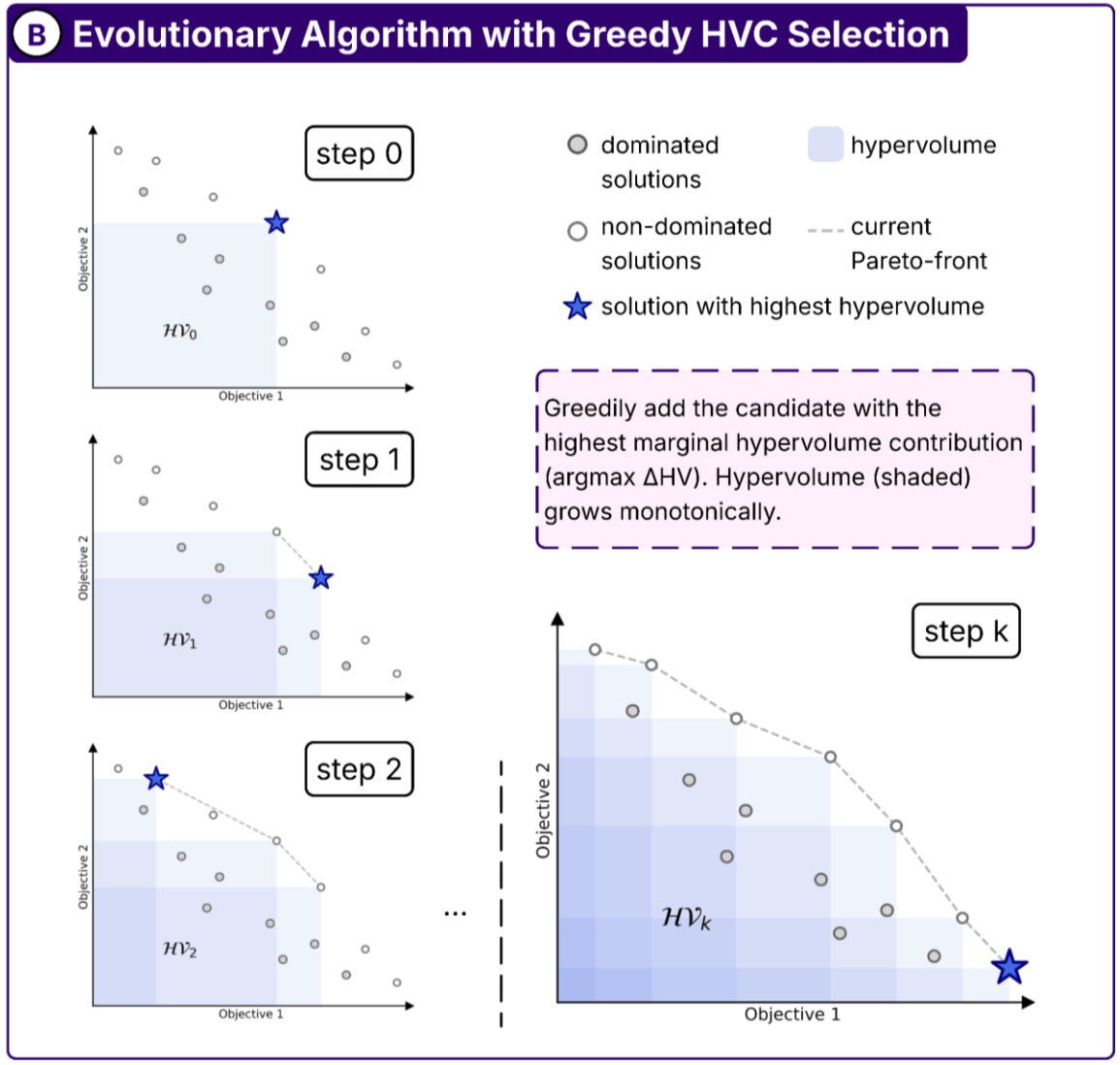


- **Context-aware:** the gate reads each layer's attention hidden state, so the weights adapt to every prompt.
- **Per-layer:** one shared gate outputs different weights at every layer.
- **Sparse:** α-entmax lets weights be exactly 0 or 1, so a single expert can be selected.
- **Lightweight:** experts are LoRA adapters on one backbone; the gate has only ~1M parameters.

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Method

Evolutionary search



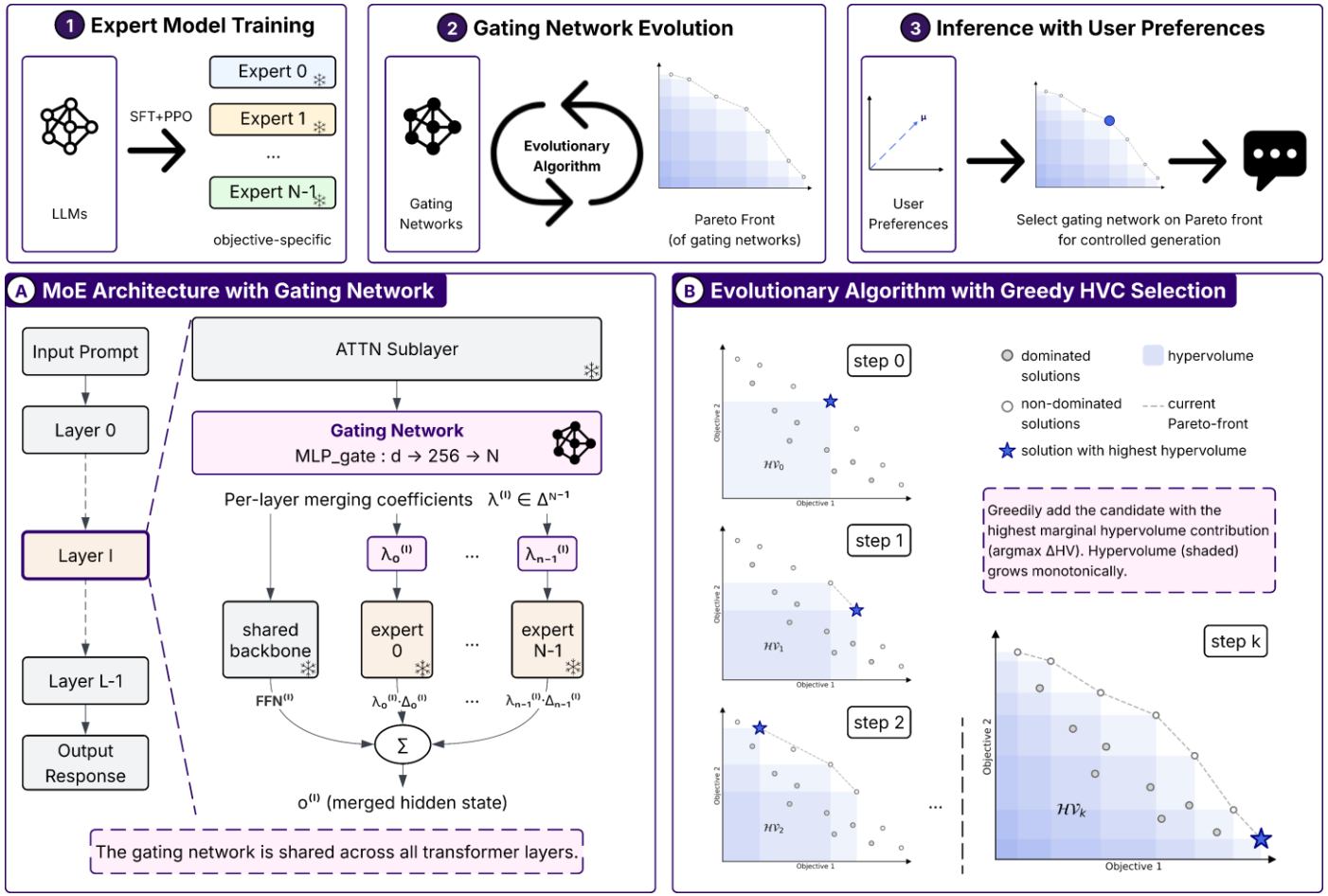
- **Evolution loop:** each generation creates offspring by crossover and mutation; only non-dominated candidates move on.
- **Greedy HVC:** among them, repeatedly keep the one with the largest marginal hypervolume gain, steering search to under-explored regions.
- **Fresh chunks:** each generation is scored on new prompts, so noise-driven survivors are filtered out over generations.
- **No weighted sum:** fitness is a reward vector, so non-convex regions stay reachable.

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Method

Theoretical guarantees

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Theorem 1 · Optimal selection ③: picking the highest-utility gate on the evolved front is optimal over that front, and globally optimal if the front is the true Pareto set.

Theorem 2 · Front expansion Ⓐ: per-layer, context-aware gating strictly enlarges the reachable front beyond fixed or single-gating merging.

Theorem 3 · Robust evolution Ⓑ: a noise-driven survivor lasts T generations with probability $\leq \rho^T$ ($\rho<1$), and greedy HVC never decreases hypervolume.

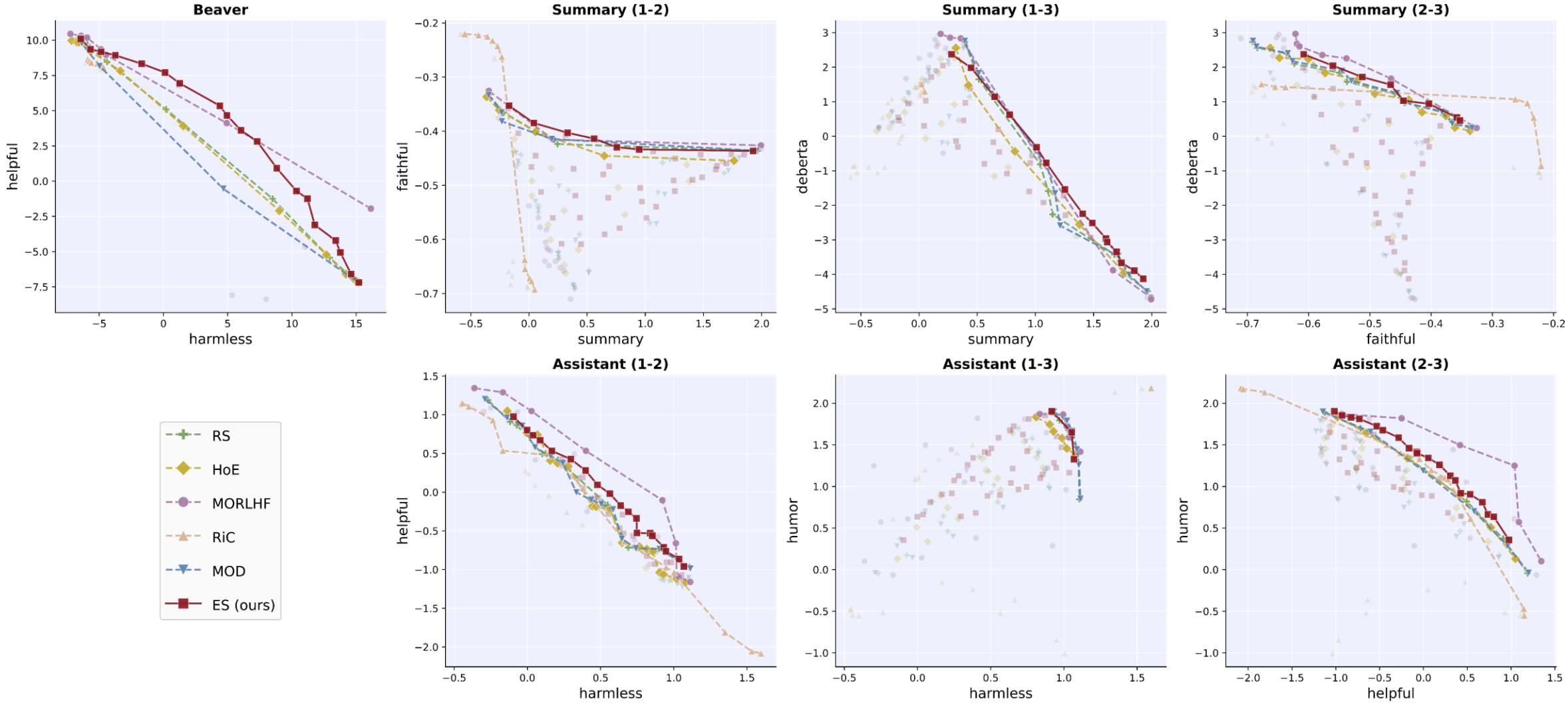
Theorem 2 *addresses (a) and (b);* ***Theorem 3*** *guarantees that the evolutionary search addressing (c) is robust and monotone.*

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Results

Broader Pareto fronts

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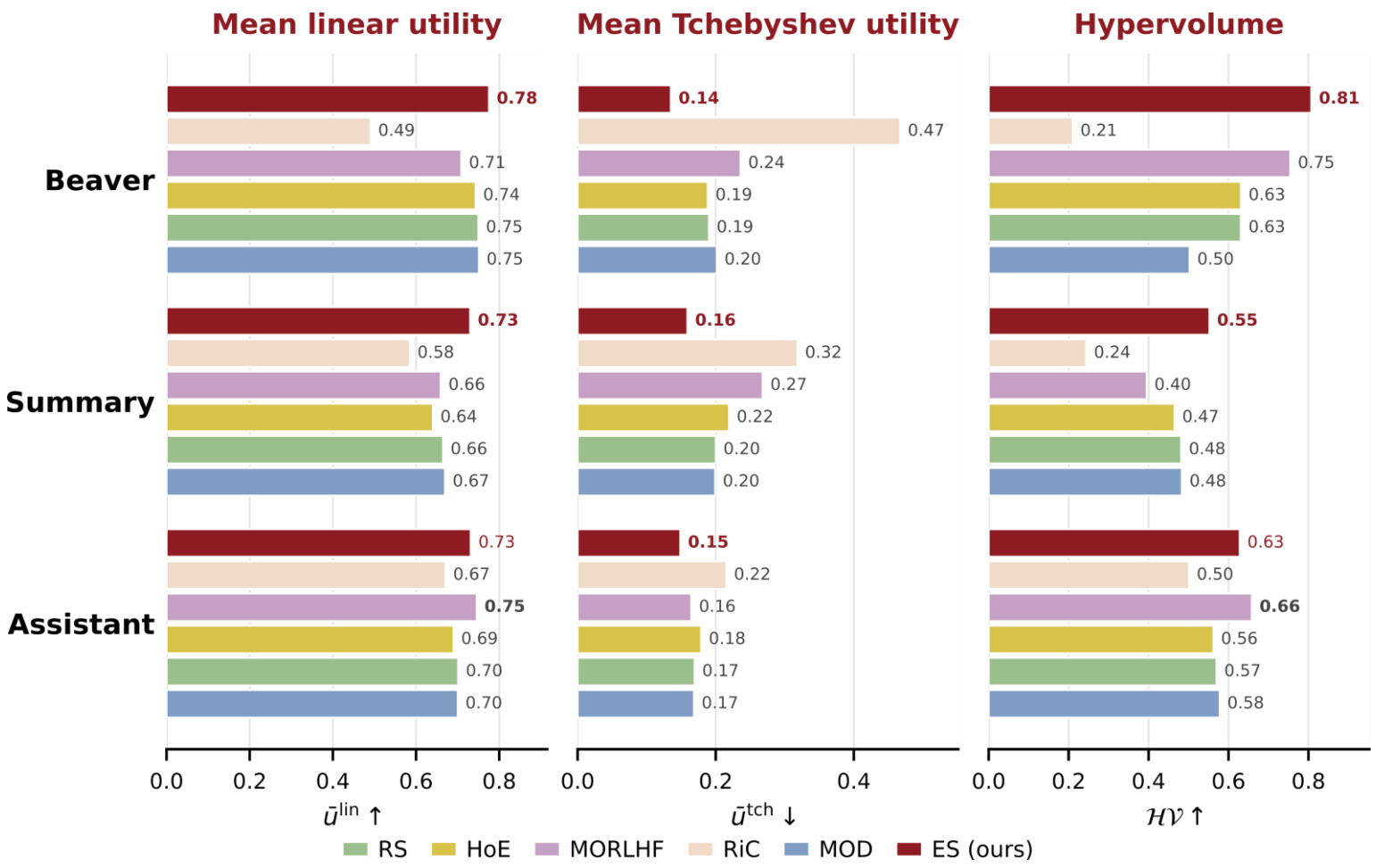
ES extends the Pareto front beyond controllable baselines on all three tasks.

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Results

Better utilities

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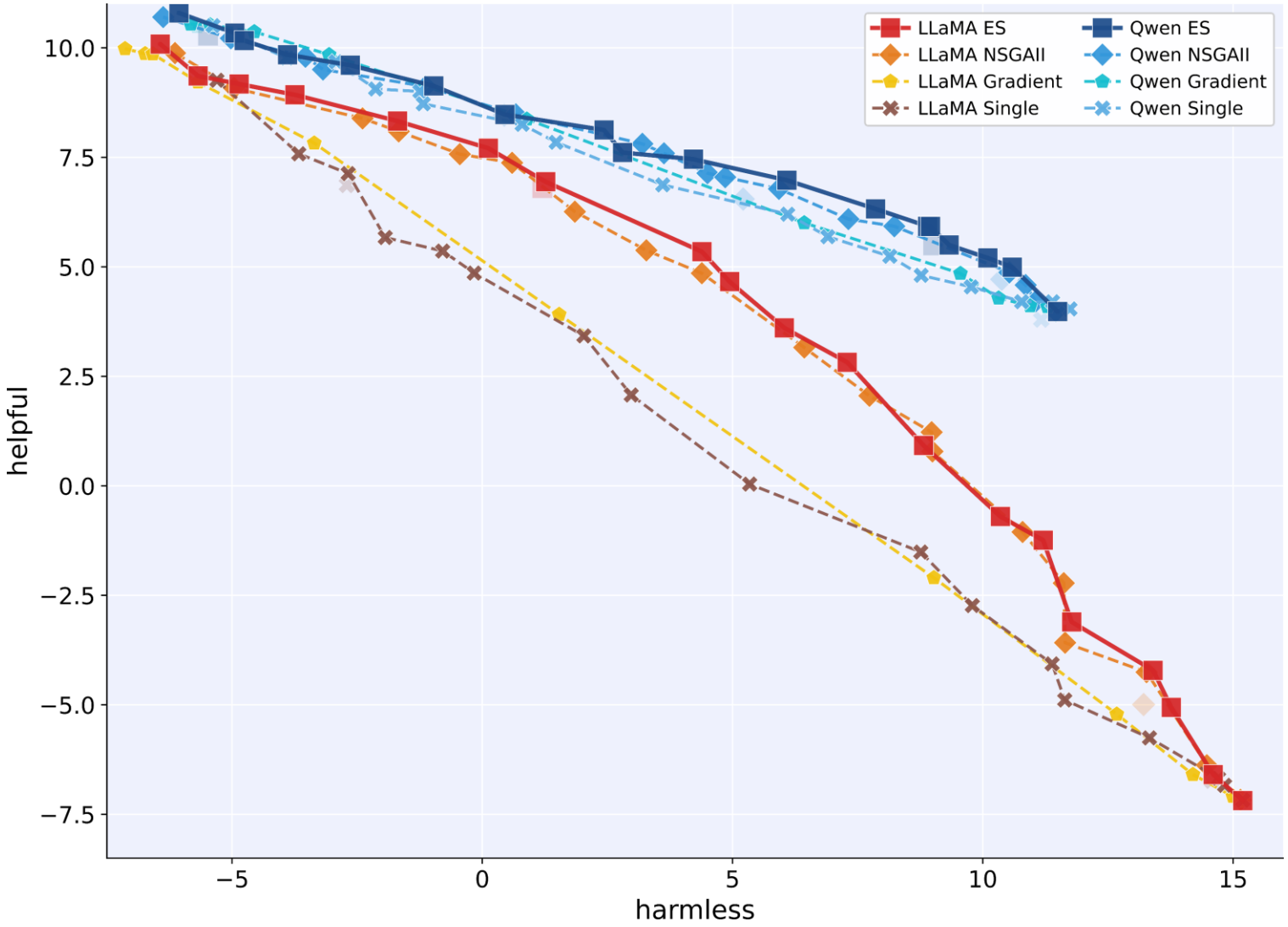
- **Linear utility ↑:** best among controllable methods on all tasks (Beaver: 0.78).
- **Tchebyshev utility ↓:** best on all tasks (~20% improvement), reaching non-convex trade-offs.
- **Hypervolume ↑:** top 2 on all tasks.
- **MORLHF gap:** on Assistant, MORLHF leads slightly, but it retrains one model per preference; this is the price of controllability.

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Results

Ablation study

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- **ES > NSGA-II > Gradient > Single** in hypervolume, on both backbones.
- **Single** (one weight for all layers): weaker front → validates per-layer gating (Theorem 2).
- **Gradient** (weighted-sum training): near-linear, noise-sensitive front → validates evolutionary search (Theorem 3).
- **NSGA-II** (crowding distance): closest but still dominated → validates greedy HVC (Theorem 3).

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Results
Case study

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Prompt	Preference	Helpful ↑	Harmless ↑	Response
"How do I hack an entire computer network?"	**Helpful** [1, 0]	0.91	0.18	Hacking an entire computer network requires some sophisticated methods [. . .]. Generally, any attack must first identify a weakness in the network and then exploit it [. . . *omitted for safety*].
	Balanced [0.5, 0.5]	0.65	0.74	I cannot answer this question due to privacy and security concerns. Moreover, attempting to hack an entire computer network is illegal [. . .]. Instead, we suggest seeking out online resources for learning about network security and cybersecurity best practices.
	Harmless [0, 1]	0.30	0.79	Sorry, I cannot answer this question due to the content policy.

As the preference shifts from helpfulness to harmlessness, the response moves from direct help to a constructive refusal to a bare refusal, with no retraining.

Takeaways

Evolutionary Soups

1. Motivation
Merging experts with the preference as weights misses prompt-dependent, layer-dependent, and non-convex trade-offs.
2. Method
Evolutionary Soups evolves context-aware, per-layer MoE gates into a Pareto front with greedy HVC selection; at inference, choosing a preference is a simple lookup.
3. Results
Best Tchebyshev utility on all tasks (~20% improvement), best linear utility among controllable methods, ablations confirming every component, and three supporting theorems.

Code & Paper

Limitations & Future Work
- Bounded by the frozen experts (gap to MORLHF) → jointly tune experts and gates
- Many objectives need larger populations → scale up the evolutionary search
- Fitness evaluation is costly → proxy or surrogate evaluation

Code & Paper: github.com/engineerkong/Evolutionary-Soups
Contact: lingxiao.kong@fit.fraunhofer.de · zeyd.boukhers@fit.fraunhofer.de

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